

Preserved Discrimination but Impaired Interpolation in hV4 Identifies the Correctable
Distortion in Colour Vision Deficiency

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Abstract

Placeholder abstract. Replace this with your actual abstract text. This study investigated neural color representations in individuals with color vision deficiency (CVD) using functional MRI and computational modeling approaches.

Keywords: color vision deficiency, fMRI, shared response model, forward encoding model, representational similarity analysis

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Introduction

Color vision deficiency (CVD) affects approximately 8% of males and 0.5% of females of Northern European descent (Bosten, 2019) and arises from altered spectral sensitivity of the L- or M-cone photopigment, driven by X-linked opsin-gene polymorphisms (Neitz & Neitz, 2011; Deeb, 2005). In anomalous trichromats the deviant cone’s peak sensitivity is displaced by approximately 2–12 nm from its normal value (Stockman & Sharpe, 2000). This shift reduces the L–M cone-opponent signal rather than abolishing it. The peripheral consequences — reduced red–green discrimination, category confusions, and elevated just-noticeable differences along the L–M axis — are the classical phenotype captured by the Ishihara test (Ishihara, 1917) and by subsequent psychophysics (Bosten, 2019).

CVD is not an on/off loss: downstream cortical color machinery remains intact and receives a quantitatively reweighted but qualitatively similar input. This makes CVD a paradigmatic example of a condition in which a high-dimensional neural representation is not absent but *structurally distorted*. The scientific and translational question is therefore not whether color information survives the retina — it manifestly does — but whether, and how, its cortical geometry is reshaped at the level of the individual. This question remains open, and an individualized assistive filter must answer it before it can be designed.

CVD correction filters in current use fall into two structurally related families. Hardware filters — dichroic notch lenses such as the EnChroma line — attenuate spectral bands near the L–M crossover to widen the post-receptoral L–M signal. Software filters operate at the display: Brettel–Viénot–Mollon simulation (Brettel, Viénot & Mollon, 1997) and the parameterized Machado retinal model (Machado, Oliveira & Fernandes, 2009) forward-project a stimulus onto an anomalous retina. Their Daltonization inverses, including recent learned reformulations (Akalın et al., 2025), redistribute the lost contrast into channels the user can still see. Despite this

diversity, the family shares two structural commitments: optimization against a *population-average* retinal model, with the same transformation applied to every user.

Attempts to escape this average-user ceiling have predominantly taken one route: tuning a retinal-model parameter — the simulated L- or M-cone shift in Machado-class models — to a user’s responses on plate-style or appearance-matching tests. The route is constrained at its target: its optimization criterion is the user’s appearance *report*, not the cortical representation that downstream tasks read out.

This distinction matters because long-term cortical adaptation in anomalous trichromats reshapes hue appearance independently of the retina (Tregillus et al., 2021). Appearance-matched calibration therefore inherits whatever compensation the cortex has already imposed. A device tuned to undo only the retinal component then re-renders the input *as if* that cortical compensation did not exist. Consistent with this account, clinical validation of consumer color-enhancing CVD-correction glasses reports at most modest changes in chromatic discrimination thresholds in congenital red–green deficiencies (Patterson et al., 2022), and recent behavioral work likewise shows that population-average filters change the appearance of color without substantively shifting discrimination thresholds at threshold contrast (Somers, Franklin & Bosten, 2024), while the underlying CVD phenotype varies markedly even within a single diagnostic category (Bosten, 2019). What is missing from this landscape is a filter derived from, and validated on, the *user’s own* neural color geometry.

Cortical color representation is a distributed, hierarchical computation (Gegenfurtner, 2003; Shapley & Hawken, 2011; Conway, Eskew, Martin & Stockman, 2018). Early visual cortex carries cone-opponent signals. Higher-order regions, in particular hV4 (Brouwer & Heeger, 2009; Bannert & Bartels, 2018), support a population code for continuous hue that can be read out by forward encoding models (Brouwer & Heeger, 2009; Parkes, Marsman, Oxley, Goulermas & Wuerger, 2009; Kuriki, Sun, Ueno, Tanaka & Cheng, 2015). Cross-subject shared-response modelling has recently been validated for healthy color decoding (Bannert & Bartels, 2025), providing a methodological scaffold that the present clinical extension adopts.

Behavioral work converges on the conclusion that anomalous trichromats do not merely discriminate less well; they compensate. Long-term cortical adaptation reshapes hue appearance (Tregillus et al., 2021; Boehm, MacLeod & Bosten, 2014; Webster, 2015), and hue-scaling experiments in anomalous trichromats ($n = 10$ AT vs. 26 NT) reveal a consistent 21.4° rotation of the blue–yellow response peak — a phase shift of the half-rectified cosine fit to scaling responses across the 360° stimulus circle (NT peak 127.5° vs. AT peak 106.1° ; $t(34) = 5.95$, $p < 0.001$) (Emery, Volbrecht, Peterzell & Webster, 2021; Isherwood, Joyce, Parthasarathy & Webster, 2020). These results locate the key variance not at the retina alone but at the retinal–cortical interface.

Yet three concrete gaps remain. (*Gap 1*) Existing neural studies of color characterize population-mean geometry in healthy observers (Brouwer & Heeger, 2009; Bannert & Bartels, 2018; Kuriki et al., 2015) or, in CVD, group means at the level of activation magnitude. To our knowledge — based on PubMed and Google Scholar searches combining “CVD” / “color vision deficiency” / “anomalous trichromacy” with “fMRI”, “forward encoding”, and “individual differences” — none has inverted the high-dimensional cortical signal into an interpretable, individualized distortion field. (*Gap 2*) CVD neural work has relied principally on classification accuracy, which is a sufficient statistic for category identity but discards the continuous relational structure (Brouwer & Heeger, 2013) that downstream perceptual and translational tasks actually demand. (*Gap 3*) Behavioral evidence for cortical compensation has not been translated into a concrete criterion for choosing between retinal-level and cortical-level correction strategies — an omission that directly limits filter design.

A filter for a continuous-hue condition must operate on continuous-hue representation, not on categorical labels. This observation motivates a complementary pair of cross-validation schemes on the same forward encoding model. Leave-one-run-out (LORO) cross-validation probes whether the 8 trained hues can be *classified* from held-out runs. It measures category-level discriminability and is the minimum requirement for a filter to matter: if the target cortex fails to discriminate the displayed colors, no stimulus-space adjustment has leverage on which to act.

Leave-one-color-out (LOCO) cross-validation, by contrast, probes whether the encoder *interpolates* from its 7 training hues to a held-out hue. It measures the integrity of the continuous manifold on which any hue correction must act.

A joint pattern is therefore predicted and empirically required: LORO *preserved* in CVD (the filter precondition) together with LOCO *impaired* (the filter target). Finding only the first would make a filter redundant; finding only the second would preclude any stimulus-space remedy. Finding both together defines the regime in which individualized corrective filters are simultaneously warranted and tractable, and in which the per-hue LOCO vulnerability vector — the subject-specific signature of where the continuous geometry fails — becomes the phenotype to be inverted.

We test whether *individual hV4 color geometry* carries person-specific distortion through a single-case fMRI characterization of two CVD adults (one moderate deutan, one moderate-severe protan) under Crawford–Howell single-case inference (see Methods), alongside a normative reference of seven healthy controls. Findings specific to sub-09 — the single protan participant in our cohort — are reported as exploratory single-case observations requiring independent replication.

1. Does the continuous cortical hue geometry in CVD carry

individually-structured distortion? We use the LORO/LOCO dissociation on hV4 forward encoding to distinguish preserved discrimination from impaired interpolation, and extract a per-hue vulnerability vector $\mathbf{v} \in \mathbb{R}^8$ as each individual’s geometric fingerprint.

2. Can that distortion be expressed in a small number of physiologically interpretable parameters, and does an independent geometric criterion corroborate the fit?

We infer 1–2 DOF parameters from $\mathbf{v}$ under three mechanistic model families (retinal Machado, retinal–cortical R+C, cortical 2-component) with closed testing. We then cross-check the winning fit against two independent signatures. The first is the ΔRDM , the pairwise difference of representational dissimilarity matrices (RDMs) in the common space derived by the Shared Response Model (SRM; Chen et al., 2015). The second is the

cross-cohort S-cone compensation angle described above.

- 3. Does the pre-image of the fitted distortion constitute a testable corrective filter, and do retinal and cortical hypotheses agree on what to display?** We compute the per-hue pre-image of each fit and assess bijectivity. Retinal and cortical fits converge on *who* is distorted but diverge on *how* to correct — generating a falsifiable behavioral prediction to be tested in a same-day 2AFC discrimination experiment.

We hypothesized that (i) categorical hue discrimination at hV4 is preserved in both CVD individuals while continuous-hue interpolation is selectively impaired along S-cone-intermediate hues; (ii) the resulting per-hue vulnerability profile is captured by a small number of physiologically interpretable parameters anchored on the S-cone opponent axis; and (iii) the pre-image of these parameters yields a stimulus-space correction filter whose direction differs between retinal-level and cortical-level model fits. Together, the three steps yield a per-individual filter whose input (cortical signal), intermediate parameters, and output (stimulus shift) are each testable in a single CVD user. Per-subject estimates are validated against an HC leave-one-out distribution under the same loss (Supplementary); case-level distinctness rests on joint anchors — HC parameter-norm bracket, directional consistency with cone-physiology, and (pending) Phase-3 behavioral validation.

Methods

The analysis follows three stages. First, the fMRI neural signature of each CVD participant’s hue-processing deficit was characterized by two measures, both serving as fitting criteria for the distortion model. Leave-one-color-out (LOCO) decoding at hV4 yielded a hue vulnerability profile $\mathbf{v} \in [0, 1]^8$ (see the Evaluating color interpolation with cross-color decoding section). The between-group difference of representational dissimilarity matrices (RDMs), denoted ΔRDM , quantified the per-pair deviation of CVD geometry from the HC mean (see the Between-group representational dissimilarity section). Second, a two-component cortical distortion model fits these neural

observations by simulating the joint effect of a retinal cone shift and a cortical orientation gain on HC-trained neural representations (see the Two-component distortion model section). Third, the fitted distortion is analytically inverted to yield a stimulus-space pre-image filter that maps the distorted hue coordinates back towards the HC percept (see the Stimulus-space filter section).

Participants

Thirteen volunteers were recruited under Institutional Review Board approval (#2510/002-023) and provided written informed consent. Color vision was screened using the 14-plate Ishihara pseudoisochromatic test (Ishihara, 1917). Participants identifying ≥ 9 plates correctly were classified as healthy controls (HC; $n = 7$, 3 female, age 22.7 ± 2.5 years). Two male participants scoring below this threshold were classified as CVD: sub-08 (deuteranomalous, 5/14 plates corrected) and sub-09 (protanomalous, 7/14 plates corrected). Four additional participants were excluded for failing to complete the full scanning protocol. All CVD results are interpreted as single-case demonstrations using the Crawford and Howell (1998) modified t -test; no population-level inference is made.

Stimuli and task

Eight isoluminant chromatic stimuli were drawn from CIE $L^*a^*b^*$ space at equal angular spacing (0° – 315° , 45° steps; $L^* = 75$, chroma = 40) (Commission Internationale de L'Eclairage, 1986), spanning red, orange, yellow, green, cyan, blue, purple, and magenta (Figure 1A). A gray filler ($a^* = b^* = 0$) served as a null condition.

Stimuli were presented in a Rapid Serial Visual Presentation (RSVP) design to sustain attention and fixation. A circular patch (1.5 s per stimulus) was shown in pseudo-random order optimized with Neurodesign (Durnez, Moerkerke & Nichols, 2018). Participants pressed a button upon detecting two consecutive 'K's within a rapid serial letter stream presented above the grating patch. Each run contained eight repetitions of all eight chromatic stimuli and the gray filler; inter-stimulus intervals were

drawn from $\{3, 4.5, 6\}$ s. Each session comprised six runs (≈ 7 min each). The experiment was coded in PsychoPy 2022.2.5 (Peirce et al., 2019).

CVD participants additionally completed a second scanning session for filter evaluation (see the Behavioral concordance section).

MRI acquisition and preprocessing

Data were acquired on a Siemens 3T MAGNETOM Cima.X scanner with BioMatrix 3T coils. To constrain coverage to posterior occipital cortex we used 24 oblique slices perpendicular to the calcarine sulcus (Brouwer & Heeger, 2009) (TR 1.5 s, TE 30 ms, FA 75° , $2 \times 2 \times 2$ mm, 96×80 matrix).

Data were converted to BIDS format and defaced with ezBIDS (Levitas et al., 2024). Functional images were registered to the T1w image via mutual-information coregistration (FreeSurfer `mri_coreg`; Maes, Collignon, Vandermeulen, Marchal & Suetens, 1997; Wells, Viola, Atsumi, Nakajima & Kikinis, 1996). T1w-to-MNI normalization used 12-DOF affine (FSL FLIRT; Jenkinson, Bannister, Brady & Smith, 2002) followed by nonlinear warping (FNIRT; Andersson, Jenkinson & Smith, 2007), yielding 2 mm isotropic MNI-space BOLD data.

ROI definition and response estimation

Bilateral V1, V2, V3, and hV4 were defined from the Wang probabilistic atlas (Wang, Mruczek, Arcaro & Kastner, 2015) at $> 50\%$ probability, intersected with each participant’s BOLD brain mask (voxel counts: V1 655 ± 214 ; V2 451 ± 145 ; V3 103 ± 29 ; hV4 63 ± 22 ; mean $\pm$ SD).

Single-trial response amplitudes were estimated with a two-stage GLM (Dale, 1999; Brouwer & Heeger, 2009, 2013). First, an ROI-level hemodynamic response function (HRF) was extracted from a finite-impulse-response (FIR) model (8-TR basis set). Voxels in the top 50% of variance explained by the mean ROI HRF were retained (V1 328 ± 107 ; V2 226 ± 73 ; V3 52 ± 15 ; hV4 32 ± 11 voxels). Second, the estimated HRF was convolved with per-color design matrices and fitted by ridge regression to yield one amplitude per (voxel, color, run), organized as a $6 \times 8 \times V$ array per subject.

Voxel amplitudes were aligned across runs using within-subject Procrustes rotation (Gower, 1975): runs 2–6 were mapped to run 1 via the orthogonal transformation minimizing $\|X_{\text{ref}} - XQ\|_F$ (scaling prohibited).

Shared Response Model

Cross-subject alignment and dimensionality reduction were performed with the Shared Response Model (SRM; Chen et al., 2015). SRM estimates per-participant orthonormal mappings $W_i \in \mathbb{R}^{V \times k}$ (V voxels, k latent dimensions) and a group-shared response matrix $S \in \mathbb{R}^{k \times 8}$:

$$\min_{W_i, S} \sum_{i=1}^N \|X_i - W_i S\|_F^2 \quad \text{s.t.} \quad W_i^\top W_i = I_k \quad (1)$$

SRM was trained exclusively on the seven HC participants to avoid circular inference in CVD comparisons. Each CVD participant was subsequently projected into the fixed HC-derived space by solving $W_{\text{CVD}} = UV^\top$ where $U\Sigma V^\top = \text{SVD}(X_{\text{CVD}} \cdot S^\dagger)$. Reduced dimensions were chosen by mean-rank aggregation across seven-fold leave-one-subject-out (LOSO) cross-validation, using three SRM-quality metrics (RDM split-half reliability, cross-subject RDM correlation, and reconstruction error): V1 $k = 4$; V2 $k = 4$; V3 $k = 3$; hV4 $k = 3$.

Forward encoding model

Both discrimination (LORO) and interpolation (LOCO) decoding used the same forward encoding model (Brouwer & Heeger, 2009). The model represents each stimulus as a $F = 6$ -dimensional channel vector $\mathbf{c}(\theta) = [c_1(\theta), \dots, c_F(\theta)]^\top$, where $c_k(\theta) = \max(0, \cos(\theta - \theta_k))^2$ is a half-wave rectified squared cosine tuned to preferred hue θ_k (preferred hues spaced equally at 60° ; Brouwer & Heeger, 2009). A weight matrix $W \in \mathbb{R}^{F \times V}$ is estimated by ridge regression with the regularization parameter α selected by generalized cross-validation (GCV; Golub, Heath & Wahba, 1979; see Supplementary for implementation details) to predict voxel amplitudes from channel activations: $\hat{X} = CW^\top$. Decoding inverts the encoding direction: for a held-out voxel

pattern $\mathbf{x}$, channel activations are reconstructed as $\hat{\mathbf{c}} = W\mathbf{x}$, and the candidate hue whose basis vector has the highest Pearson correlation with $\hat{\mathbf{c}}$ is returned as the decoded color (Figure 2). Alternative decoders evaluated during model selection are described in Appendix .

Evaluating color discrimination with cross-run decoding

Categorical discrimination was assessed with leave-one-run-out (LORO) cross-validation in SRM-aligned space using the forward encoding model (see the Forward encoding model section). For each of the six runs, W was estimated on the remaining five runs using mean-subtracted SRM projections, and each held-out run’s 8 color patterns were decoded by nearest-neighbour matching. Primary discrimination performance was the 8-class exact accuracy (chance = $1/8 = 0.125$), averaged across six folds.

Group-level HC–CVD comparison used a Mann–Whitney U test on cross-subject generalization scores: the encoding model trained on each HC participant’s SRM-aligned data was applied to every other participant’s aligned patterns, yielding 21 unique HC-to-HC and 14 HC-to-CVD pairwise generalization scores. Within-ROI single-case tests used the Crawford and Howell (1998) modified t -test (one-tailed, lower).

Evaluating color interpolation with cross-color decoding

Continuous hue interpolation was assessed with leave-one-color-out (LOCO) cross-validation using the same forward encoding model (see the Forward encoding model section). Each of the 8 hues was held out in turn; W was estimated on the remaining 7 hues using all 6 runs (pooled 42-sample design matrix), and the held-out hue was predicted by nearest-neighbour matching in channel-output space.

Adjacent accuracy. The primary cross-color decoding metric was adjacent accuracy: the proportion of predictions that fell within ± 1 hue step (45°) of the target hue, averaged across the 6 runs of each test color (range 0–1; chance level under a uniform random predictor = $3/8 = 0.375$; exact-accuracy chance = $1/8 = 0.125$). For HC above-chance testing and for pairwise HC–CVD comparisons at hV4, p -values were

obtained from 1,000 random color-label permutations and from the Crawford and Howell (1998) modified t -test (one-tailed, lower), respectively.

Vulnerability profile. The 8-dimensional vector of per-hue adjacent-accuracy values, $\mathbf{v} \in [0, 1]^8$, constitutes each participant’s *hue vulnerability profile*. For group-level model fit, we report Spearman ρ between the distortion-model-predicted and observed $\mathbf{v}$ (see the Two-component distortion model section). Per-hue group comparisons (Figure 3C) used the Crawford and Howell (1998) modified t -test (one-tailed, uncorrected). Effect sizes are reported as Hedges’ d (Hedges & Olkin, 1985) with the Crawford–Howell correction for small- n comparisons.

Between-group representational dissimilarity

To characterize the representational geometry of LOCO impairment in SRM-aligned space, we computed two measures. Mean pairwise disparity captures whether CVD geometry is globally more dispersed than HC. The between-group RDM difference, ΔRDM , captures which color pairs drive the deviation and tests whether a cortical distortion model accounts for the pattern.

Pairwise disparity. For each participant and ROI, the RDM was the 8×8 matrix of pairwise correlation distances ($1 - \text{Pearson } r$). Mean pairwise disparity was the average of the 28 upper-triangle elements. To provide an unbiased HC reference for the held-out HC participant, the HC disparity distribution was estimated by a leave-one-subject-out (LOO) procedure. Under the directional prediction that CVD geometry is more dispersed than HC, individual CVD disparity was compared to the HC LOO distribution by the Crawford and Howell (1998) modified t -test (one-tailed, upper). Significance was assessed at $\alpha = 0.05$.

Geometric deviation from HC and model correspondence. The per-pair deviation of each CVD participant’s RDM from the HC mean was $\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC}}$, where $\overline{\text{RDM}}_{\text{HC}}$ is the grand mean RDM over the seven HC participants. Correspondence between the observed ΔRDM and the prediction of the two-component distortion model (see Two-component distortion model) was

reported descriptively as the Spearman ρ between the upper-triangle elements of the two matrices.

Two-component distortion model

CVD hue distortion was modelled as a two-component angular shift in Stockman opponent-color space:

$$\delta\theta(h; \beta_s, \beta_c) = \beta_s \cos(h - 90^\circ) + \beta_c \cos(h - \theta_{\text{conf}}) \quad (2)$$

where h is the opponent-hue angle of the stimulus in CIELab space and θ_{conf} is the CVD-subtype confusion axis (16° protan; 150° deutan), derived from the Brettel–Viénot–Mollon dichromatic simulation (Brettel et al., 1997) applied to Stockman & Sharpe LMS cone fundamentals (Stockman & Sharpe, 2000). β_s parameterizes distortion along the S-cone axis ($h = 90^\circ$), motivated by cortical-RDM evidence that CVD errors concentrate at S-cone loci (Emery et al., 2021). β_c parameterizes rotation along the confusion axis, with both signs unconstrained as free parameters of the fit. We compared this model with retinal-shift-based alternatives (Machado, R+C retinal family); see Appendix .

Per-subject parameters (β_s, β_c) were recovered by minimizing a four-term, neural-primary loss derived from the V4 LOCO forward model:

$$L_{\text{fit}}(\beta_s, \beta_c) = 1.0 L_{\text{vuln}} + 0.5 L_{\text{rank}} + 0.2 L_{\text{rdm}} + 0.1 L_{\text{smooth}} \quad (3)$$

with each term normalized to $[0, 1]$ before weighting. Let $\mathbf{v}_{\text{sim}}(\beta_s, \beta_c) = f_{\text{forward}}(C(\theta + \delta\theta(\cdot; \beta_s, \beta_c)); W) \in [0, 1]^8$ be the simulated per-color V4 LOCO vulnerability ($1 - \rho$) obtained by re-applying the HC-trained ridge encoding weights W (Section) to the design matrix C shifted by the candidate distortion, and let $\mathbf{v}_{\text{obs}} \in [0, 1]^8$ be the observed CVD per-color vulnerability.

Loss-term roles. $L_{\text{vuln}} = \frac{1}{8.4} \|\mathbf{v}_{\text{sim}} - \mathbf{v}_{\text{obs}}\|^2$ (per-color MSE, normalizer 4.0 for $[0, 1]$ amplitude range) is the primary fit term: it forces the simulated vulnerability profile to match the per-color LOCO breakdown that the CVD encoder exhibits, which

is the operative neural quantity for the corrective filter (see the Discussion section for the LOCO-behavior concordance evidence underlying this choice).

$L_{\text{rank}} = (1 - \rho_{\text{Spearman}}(\mathbf{v}_{\text{sim}}, \mathbf{v}_{\text{obs}}))/2$ is an ordering term that breaks ties when the MSE landscape is locally flat by penalizing rank-reordering of vulnerable colors; it is downweighted to 0.5 because amplitude (MSE) is the operative signal.

$L_{\text{rdm}} = (1 - \cos(\Delta\text{RDM}_{\text{sim}}, \Delta\text{RDM}_{\text{obs}}))/2$ is a geometric convergence check on the 28-pair upper-triangle of the CVD-minus-HC representational dissimilarity-matrix difference at hV4 ($\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC}}$, correlation distance, Kriegeskorte, Mur & Bandettini, 2008). It is downweighted to 0.2 because the RDM and per-color vulnerability signals partially overlap. $L_{\text{smooth}} = \frac{1}{8.32400} \sum_c (\delta\theta_{c+1} - \delta\theta_c \bmod \pm 180)^2$ is a circular-smoothness regularizer on the per-color shift vector $\delta\theta \in \mathbb{R}^8$ produced by the candidate (β_s, β_c) ; it is downweighted to 0.1 as a weak prior favouring smooth filters over discontinuous per-color shifts.

ROI scope. All four terms are computed at hV4 (V4 on disk). We fit at hV4 rather than V1–V3 because hV4 is the only ROI in this dataset whose HC group LOCO interpolation exceeds the permutation null ($p = 0.044$ at hV4 vs. NS at V1/V2/V3; Color interpolation is impaired in both CVD cases), and because hV4 is the canonical cortical hub for hue representation in the ventral stream (Brouwer & Heeger, 2009; Bannert & Bartels, 2018; Kuriki et al., 2015); cone-shift parameters fit on an encoder that cannot interpolate held-out colors lack a voxel-level cone-shift interpretation.

Per-subject distortion fit. Parameters were searched by exhaustive grid over $\beta_s \in [0^\circ, 50^\circ]$ and $\beta_c \in [-50^\circ, 50^\circ]$ ($26 \times 51 = 1,326$ cells at 2° resolution); $(\hat{\beta}_s, \hat{\beta}_c) = \arg \min L_{\text{fit}}$ was retained per CVD subject. We checked the optimal distortion magnitude against an HC leave-one-out distribution under the same loss for validation (Supplementary); quantitative filter validity is established by the Phase-3 2AFC behavioral arm (see the Behavioral concordance section, last paragraph).

Stimulus-space filter

The subject-specific correction filter was derived as the pre-image of the fitted 2-component transform $T(\hat{\beta}_s, \hat{\beta}_c) : \theta \mapsto \theta + \delta\theta(\theta; \hat{\beta}_s, \hat{\beta}_c)$. For each of the 8 displayed hues θ_k , we solved for the input angle $\tilde{\theta}_k$ such that $T(\tilde{\theta}_k) = \theta_k$, yielding a per-hue correction $\delta\theta_k^{\text{filt}} = \tilde{\theta}_k - \theta_k$. A CVD observer viewing $\tilde{\theta}_k$ thus receives, under the model, the same cortical hue representation as a healthy control observer viewing θ_k .

Pre-images were computed by numerical inversion of the forward map (Brent’s-method root-finding); inversion was exact ($|T(\tilde{\theta}_k) - \theta_k| < 0.001^\circ$) for all 8 hues in both CVD participants.

Behavioral concordance

Behavioral tasks were held on the same day as the MRI acquisition.

Just-noticeable difference (JND). Color discrimination thresholds were estimated for eight hue pairs using two-interleaved 1-up/1-down adaptive staircases, each terminating after eight reversals (Levitt, 1971). Pairs spanned three categories: (i) three pairs hypothesized as universally difficult for both CVD types (yellow–purple, blue–purple, red–orange), (ii) two pairs hypothesized as deutan-specific (orange–yellow, yellow–green), (iii) two pairs hypothesized as protan-specific (cyan–magenta, green–blue), and (iv) one control pair (red–cyan).

For each pair, the CVD JND threshold was compared to the HC distribution and assigned a behavioral impairment label: *hypo* when the CVD threshold fell above the HC range (CVD/HC ratio > 1 ; discrimination worse than HC), or *hyper* when it fell below (CVD/HC ratio < 1 ; discrimination preserved or better than HC). This pairwise hypo/hyper classification quantified where CVD behavior diverged from HC.

Simulator–JND concordance. The fitted 2-component simulator was tested against the behavioral labels by comparing each pair’s hypo/hyper classification with the simulator’s predicted per-pair angular distortion under the fitted parameters $(\hat{\beta}_s, \hat{\beta}_c)$. For a pair (θ_a, θ_b) , the simulator’s predicted distortion magnitude was the sum of the per-color shifts applied to its two members, $|\delta\theta(\theta_a; \hat{\beta}_s, \hat{\beta}_c)| + |\delta\theta(\theta_b; \hat{\beta}_s, \hat{\beta}_c)|$. The

simulator and behavior were called concordant when hypo-labeled pairs corresponded to larger simulator-predicted distortions than hyper-labeled pairs, providing a direct neural-to-behavioral check independent of the fit loss.

Behavioral color identification. An eight-alternative forced-choice (8AFC) color identification task was administered: participants performed six trials per color (48 trials per participant in total), and both the order of presented colors and the order of the eight answer options were randomized across trials. In parallel to the JND-simulator concordance above, each CVD response was evaluated against the simulator’s predicted shifted angle: for a presented hue θ_k , the fitted 2-component simulator predicts that responses should cluster near $\theta_k + \delta\theta(\theta_k; \hat{\beta}_s, \hat{\beta}_c)$. Exact and adjacent-bin agreement between observed responses and this predicted angle were tabulated as post-hoc consistency checks; quantitative validation is deferred to the Phase-3 2AFC arm.

Filter evaluation session (CVD only). CVD participants completed a second scanning session after the main experiment. Three runs were acquired under the 2-component-derived stimulus-space filter and three runs under a generic Windows accessibility filter, presented in counterbalanced order. LOCO decoding and LORO classification were repeated on second-session data to assess filter efficacy.

[TODO] Planned validation. A two-alternative forced-choice (2AFC) discrimination arm targeting filter-induced perceptual restoration is pre-registered for Phase 3 (data not collected at the time of writing).

Reproducibility

All analyzes were conducted in Python 3.10 (numpy 1.24.3, scipy 1.11.3, scikit-learn 1.3.0, BrainIAK 0.11). Random seeds were fixed (`seed=42`) for all permutation tests and bootstrap resampling. Analysis code and preprocessing scripts are available at https://github.com/Transconnectome/colorBlind_analysis. Anonymised preprocessed data are available on reasonable request to the corresponding author.

Results

Color discrimination is preserved across all ROIs in both CVD cases

A corrective stimulus-space filter is valid only if the visual cortex of a CVD individual retains categorical discrimination among the target hues. We verified this precondition using leave-one-run-out (LORO) cross-validation of the forward encoding model in SRM-aligned space.

Both CVD participants exceeded the 0.125 exact-accuracy chance level at every ROI (Figure 3A). A cross-subject generalization test of the same forward encoding model (HC-to-HC vs. HC-to-CVD pair-wise cross-decoding, all ROIs pooled) showed no HC-CVD difference (Mann-Whitney U test, 21 HC-to-HC vs. 14 HC-to-CVD pairs, $p = 0.668$, rank-biserial effect size and exact test statistic in Supplementary). The same null held at the within-ROI level at hV4 (Crawford & Howell t -test, both CVD subjects within the HC range, $p = 0.142$). The 8 scanner stimuli are therefore individually distinguishable at the cortical level in both CVD participants, in line with prior work on CVD above-threshold identification (Bosten, 2019; Boehm et al., 2014). The representational substrate on which a corrective filter could act is present.

Color interpolation is impaired in both CVD cases

We next localized where continuous hue interpolation fails, which defines the functional target of the corrective transform, using leave-one-color-out (LOCO) decoding of the forward encoding model. Among the four retinotopic ROIs, only hV4 supported above-chance hue interpolation in HC (adjacent accuracy 0.47 ± 0.05 , mean $\pm$ SEM, $p = 0.044$ under $8! = 40,320$ exact label permutations). V1-V3 did not exceed chance despite preserving cross-run discrimination. This confirms hV4 as the primary interpolation gate, replicating Brouwer and Heeger (2009).

Both CVD participants fell at or below the adjacent-accuracy chance level at hV4 (Figure 3B). Sub-08 reached adjacent accuracy 0.25 (Crawford & Howell $t = -1.58$, $p = 0.082$, n.s., $d_{cc} = -1.71$). Sub-09 reached 0.13 ($t = -2.48$, $p = 0.024$, $d_{cc} = -2.68$). The impairment was not uniform across hues. Both participants showed near-zero

adjacent accuracy specifically for S-cone intermediate colors at hV4, namely blue, purple, and magenta (Figure 3C). Per-hue Crawford & Howell tests reached significance for blue (sub-08 $d = 2.13$, $p = 0.047$; sub-09 $d = 2.15$, $p = 0.046$) and for purple (sub-08 $d = 2.40$, $p = 0.033$). These per-hue accuracies form each participant’s hue vulnerability profile $\mathbf{v} \in [0, 1]^8$.

Color geometry is distorted at a distinct ROI in each CVD case

To characterize the geometric basis of the LOCO impairment, we compared the pairwise representational structure of CVD and HC participants in a shared SRM-aligned space. For each CVD participant we computed disparity and the per-pair RDM deviation from the HC mean ($\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC}}$).

The ROI specificity of elevated disparity diverges between the two CVD participants, with V2-dominant disparity in deutan and V1-dominant disparity in protan (Figure 4B). Sub-08 showed significantly elevated disparity specifically at V2 (Crawford & Howell $p = 0.040$), with no significant elevation at V1, V3, or hV4. Sub-09 showed significantly elevated disparity specifically at V1 ($p = 0.007$), with no significant elevation at V2, V3, or hV4. Full test statistics and effect sizes (t , d_{cc}) for these Crawford–Howell comparisons are reported in Supplementary .

The ΔRDM heatmaps (Figure 4A) reveal subject-specific distortion structures. Sub-08 shows elevated distances among S-cone intermediate pairs at V2, and sub-09 shows a broader reorganization of pairwise distances at V1. The ROIs of strongest disparity elevation (V1 for sub-09, V2 for sub-08) differ from the ROI used as the distortion-model loss target (hV4, see the Two-component distortion model section). Mean pairwise disparity is an existence statement about absolute geometric perturbation in early visual cortex. hV4 is the cortical stage at which the corrective filter must act, so the ΔRDM enters the loss as a geometric convergence check at that ROI. V1/V2 disparity at upstream ROIs is parallel evidence of distortion rather than a competing fit target.

A 2-component cortical distortion fits both CVD cases at hV4

To identify the distortion parameters needed for the corrective filter, we fit the 2-component cortical model (Equation 2, β_s for S-cone-axis dilation, β_c for confusion-axis rotation) to each CVD participant’s hV4 representation under the four-term neural-primary loss specified in Equation 3 (see the Two-component distortion model section). hV4 was the fit ROI because it is the only ROI whose HC group LOCO interpolation exceeds the permutation null in this dataset ($p = 0.044$ at hV4, not significant at V1/V2/V3, Color interpolation is impaired in both CVD cases). Cone-shift parameters fit at hV4 are therefore interpretable as voxel-level shifts in a working interpolation manifold.

The fit returned per-subject parameters $(\hat{\beta}_s, \hat{\beta}_c) = (38^\circ, -14^\circ)$ for sub-08 (deutan, $\|\hat{\beta}\| = 40.5^\circ$) and $(6^\circ, -22^\circ)$ for sub-09 (protan, $\|\hat{\beta}\| = 22.8^\circ$, Figure 5). Both $\hat{\beta}_c$ values are negative. The etiological asymmetry between participants does *not* reside in the confusion-axis sign but in the S-cone-axis magnitude ($\hat{\beta}_s = 38^\circ$ for sub-08 vs. 6° for sub-09). We deliberately do *not* interpret either $\hat{\beta}_c$ sign as instantiating a Brettel-style cone-physiology prediction. The Brettel–Viénot–Mollon dichromat algorithm describes spectral collapse onto a missing-cone half-plane and, in our cosine parameterization, predicts $\beta_c \approx 0$ rather than a subtype-specific polarity (see the Two-component distortion model section). The non-zero $\hat{\beta}_s$ is directionally consistent with Emery and colleagues’ (Emery et al., 2021) report of S-cone-axis dominance in CVD hV4 RDM geometry, but the two quantities are defined on different physical scales and we make no numerical comparison.

We checked the optimal distortion magnitude against an HC leave-one-out distribution under the same loss for validation (Supplementary). Sub-08’s case-level distinctness rests on the R+C diagnostic (Appendix) and on the per-color filter profile (Figure 6) rather than on $\|\hat{\beta}\|$ alone.

The model-class expressiveness has a closed-form upper bound. The magnitude $|\delta\theta|$ at $\theta = 45^\circ$ is bounded by $|0.71\hat{\beta}_s - 0.26\hat{\beta}_c|$, so corrective shifts exceeding $\sim 51^\circ$ at the deutan mid-axis (e.g. an orange-to-green excursion of $\gtrsim 85^\circ$) lie outside the

2-component class for any biologically plausible $\hat{\beta}$ and would require an additional mechanism (e.g. saturation or lightness) outside the present scope.

Per-subject stimulus-space filter

The angular form of the 2-component model is bijective by construction, so a stimulus-space pre-image $\tilde{\theta}_k$ such that $T(\tilde{\theta}_k) = \theta_k$ exists for any target hue θ_k (see the Stimulus-space filter section). The per-subject correction filter $\delta\theta_k^{\text{filt}} = \tilde{\theta}_k - \theta_k$ was computed by Brent’s-method refinement of the forward map. For both participants, 8/8 pre-images were exact (residual $< 0.001^\circ$). Per-hue filter output is visualized in Figure 6 as a 4-column rendering (*Original* $\rightarrow$ *CVD perceives* $\rightarrow$ *Filtered* $\rightarrow$ *CVD(Filterd)*).

For sub-08 (deutan, $\hat{\beta}_s = 38^\circ$, $\hat{\beta}_c = -14^\circ$, Stockman confusion axis $\theta_{\text{conf}} = 150^\circ$), the filter applies a wide-amplitude L/M-rotation. A near-uniform negative shift is applied across red, orange, yellow, green, and cyan ($\delta\theta^{\text{filt}}$ ranging -10° to -32° , peak -32° at cyan, $\theta = 180^\circ$). A large opposite-sign rotation is applied at purple ($+29^\circ$, $\theta = 270^\circ$) and magenta ($+19^\circ$, $\theta = 315^\circ$). The mean correction magnitude is $|\delta\theta^{\text{filt}}| = 22^\circ$.

For sub-09 (protan, $\hat{\beta}_s = 6^\circ$, $\hat{\beta}_c = -22^\circ$, Stockman axis $\theta_{\text{conf}} = 16^\circ$), the correction is a far smaller monotonic rotation along the protan confusion axis. Peak displacements are at blue ($+21^\circ$, $\theta = 225^\circ$) and magenta (-21° , $\theta = 315^\circ$), with near-zero shifts at green and cyan. The mean correction magnitude is $|\delta\theta^{\text{filt}}| = 10^\circ$.

The two correction signatures reflect distinct CVD etiologies rather than mirrored protan/deutan symmetry. The R+C cone-shift diagnostic recovers a near-zero retinal shift with large cortical overshoot for sub-08 ($\Delta\lambda \approx 2.5$ nm, $g = -2.25$, cortical-dominant) and a large retinal shift with near-physiological cortical compensation for sub-09 ($\Delta\lambda \approx 19.5$ nm within the published protan range, $g = -1.10$, retinal-dominant; Appendix). A single retinal-family filter cannot serve both individuals.

Same-day 8-alternative forced-choice (8AFC) color identification confirmed the cortical impairment at the behavioral level (HC 0.94 ± 0.04 , sub-08 0.71, sub-09 0.62,

chance = 0.125). Sub-08’s errors clustered on the red–green axis (c2/c3 confused with c7, mean confusion 28%), and sub-09’s errors extended across the L–M opponent range with additional magenta (c8) misidentification, mirroring the per-subject hV4 LOCO vulnerability profiles (Figure 3C). Quantitative behavioral validation of filter efficacy is deferred to the Phase-3 2AFC arm (see the Behavioral concordance section).

Discussion

In this study, we tested whether *individual hV4 color geometry* is sufficiently structured to ground a person-specific corrective filter in color vision deficiency. We found that two CVD adults retained 8-class hue discrimination at hV4 but selectively lost continuous-hue interpolation, and that the residual distortion was captured by a 2-component angular-dilation model whose pre-image was exact for all eight hues in both participants (sub-08 mean $|\delta| = 22.2^\circ$; sub-09 mean $|\delta| = 10.2^\circ$). These results suggest that the cortical site at which CVD color information is geometrically distorted is also the site at which it can be inverted into a stimulus-space correction.

The dissociation between LOCO interpolation capacity and SRM Δ RDM geometric distance — 100% vs. 33% concordance with same-day JND thresholds — identifies which cortical quantity is the operative corrective target. Representational geometry (RDM) reflects the metric structure of neural color space; LOCO interpolation reflects whether that structure supports functional prediction of withheld hues. That only LOCO predicts behavioral color confusions is consistent with Brouwer and Heeger (2009), who showed that hV4 — but not early visual areas — supports interpolation to held-out hues in typical observers. The present results extend this dissociation to these two CVD cases: the same interpolation failure that predicts JND confusion is what the correction filter must address. Geometric distance metrics alone are insufficient to identify it.

The S-cone axis component (β_s) of the fitted distortion field is structurally consistent with prior evidence for cortical color compensation in anomalous trichromats. Emery et al. (2021) reported that CVD perceptual errors concentrate near S-cone

confusion loci in behavioral hue-scaling tasks, and Tregillus et al. (2021) showed that V2/V3 BOLD responses in anomalous trichromats include a cortical gain component (g) that partially compensates the retinal cone shift. These individually structured hV4 residuals may reflect cone-class-specific differences in V2/V3 compensation gain that survive into hV4, where the incompletely cancelled L–M signal manifests as a person-specific distortion of the continuous hue code. This suggests that the present 2-component model extends earlier population-mean findings to the individual hV4 level: the residual distortion that survives V2/V3 compensation reaches hV4 as an individually structured pattern. The two CVD participants in fact present qualitatively distinct correction signatures rather than mirror-image realisations of a single mechanism. Sub-08 (deutan) exhibits a wide-amplitude L/M-rotation with large $\hat{\beta}_s = 38^\circ$ and a confusion-axis term $\hat{\beta}_c = -14^\circ$; the R+C diagnostic decomposition recovers a near-zero retinal shift ($\Delta\lambda \approx 2.5$ nm) coupled with a large cortical overshoot ($g = -2.25$), pointing to a cortical-dominant CVD etiology. Sub-09 (protan), by contrast, exhibits a far smaller monotonic rotation along the protan confusion axis (small $\hat{\beta}_s = 6^\circ$, $\hat{\beta}_c = -22^\circ$); the R+C diagnostic recovers a large retinal shift ($\Delta\lambda \approx 19.5$ nm, within the published protan cone-shift range (Tregillus et al., 2021)) and a near-physiological cortical compensation ($g = -1.10$), pointing to a retinal-dominant etiology. This cortical-vs-retinal etiological dissociation is what a population-average retinal model, applied uniformly to all members of a given CVD subtype, fails to capture: in sub-08 the residual distortion is generated downstream of the cone shift and cannot be neutralized by a spectral filter alone, while in sub-09 the etiology is retinal-dominant but the 1-DOF Machado class is structurally non-invertible at three of this participant’s hues (Appendix , §A.2), so the 2-component class is required for filter realizability in both cases.

All distortion models — 1-DOF Machado, 2-DOF R+C, and 2-component — identify the same two participants as requiring correction, but their prescribed correction vectors diverge in direction (Appendix). This detection–correction dissociation constitutes the study’s primary falsifiable prediction. In a 2AFC

experiment comparing the 2-component and retinal-family filter conditions for each participant’s most LOCO-vulnerable hue pairs, only the cortically-grounded 2-component filter should significantly reduce JND. If the retinal-family filter matches or exceeds the 2-component filter in reducing discrimination thresholds, the cortical distortion account would require revision. This preregistered comparison directly tests whether cortical or retinal distortion is the corrective target for CVD color correction.

An alternative reading is that the hV4 LOCO impairment merely reflects reduced V1–V3 input variance rather than cortex-level distortion — that the bottleneck is upstream signal quality and hV4 inherits its deficit passively. The preserved LORO accuracy at hV4 argues against this account: classification and interpolation read out from the same upstream cortical input, so a generic input-variance reduction should degrade both. The observed dissociation instead localises the distortion to the continuous-hue manifold readout, consistent with hV4 as the cortical site of integration over cone-opponent signals (Brouwer & Heeger, 2009; Bannert & Bartels, 2018).

Three considerations bound the proof-of-concept scope of these findings. First, the CVD sample is $N = 2$; single-case statistics (Crawford & Howell, 1998) permit individual-level inference but population-level claims about deutan or protan subtypes require replication with larger samples. Parameter-level bootstrap confidence intervals on $\hat{\beta}_s$ and $\hat{\beta}_c$ were not computed; the inter-subject asymmetry in $\hat{\beta}_s$ (38° vs. 6°) may reflect a robust subtype-level feature, an individual-level idiosyncrasy, or both, and power-adequate replication is required to distinguish these possibilities. The *2-component* label should not be read as confirming a two-mechanism account for both participants. Second, the corrective filter is a stimulus-space prediction derived from a within-subject neural fit; filter validity is therefore a behavioral question (whether each $\tilde{\theta}_k$ restores the corresponding HC percept), not a within-cohort statistical one. We deliberately do not report fit-landscape permutation p -values for the per-subject $(\hat{\beta}_s, \hat{\beta}_c)$, because such tests address only whether a non-trivial minimum exists on a participant’s vulnerability profile and do not constrain whether the inverted filter restores HC-equivalent perception. Quantitative filter validation is provided by the

pre-registered Phase-3 2AFC discrimination arm, which compares pre-image filter, retinal-family filter, and no-filter conditions on each participant’s most LOCO-vulnerable hue pairs. Third, the effective HC pool ($n = 6$ at hV4 after excluding sub-07 for low voxel count) is small enough that the HC $\|\hat{\beta}\|$ leave-one-out distribution we report is a magnitude anchor for the CVD estimates, not a hypothesis test, and external cross-cohort hue-scaling data would be required to arbitrate population-level distortion patterns beyond this proof-of-concept sample.

This pipeline constitutes a proof-of-concept that *individual hV4 color geometry*, measured via fMRI LOCO decoding and inverted through an angular-dilation model, can ground a color correction that is architecturally distinct from population-average retinal approaches. Where Machado-type correction applies a fixed spectral shift to all individuals of a given CVD subtype, the present approach encodes each individual’s hV4 distortion field and inverts it exactly — without assuming population homogeneity. Phase 3 will test whether this cortical-level correction reduces discrimination thresholds where retinal-level correction does not.

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Appendix A: Alternative Model Comparison

The main text adopts a 2-component cortical model (Section , fit under the V4 LOCO neural-primary loss L_{fit} of Equation 3). Here we report two cone-shift baselines as reference comparisons:

- **Machado (1-DOF)** — a single retinal cone-spectral shift $\Delta\lambda$ Machado et al., 2009.
- **Retinal + Cortical (R+C, 2-DOF)** — retinal shift $\Delta\lambda$ plus a cortical compensation gain g Tregillus et al., 2021.

A.1 Fit results (LOCO Spearman ρ argmax, historical reference)

Under the LOCO- ρ argmax criterion used in earlier drafts, the Machado model fit Sub-09 at $\Delta\lambda = 13.5$ nm with $\rho_{V4} = 0.76$ and Sub-08 at $\rho_{V4} = 0.62$ (lower fit quality); the R+C model did not improve on either single-component model in either subject. These fits are reported as historical reference for the main-text 2-component fit (Section); the present L_{fit} loss (Equation 3) and LOCO- ρ argmax need not agree, since L_{fit} optimizes a combination of per-color V4 vulnerability MSE, Spearman rank ordering, ΔRDM cosine, and a smoothness regularizer rather than the rank-order monotonicity of ρ at hV4 alone.

R+C decomposition as a diagnostic. Although the R+C 2-stage model is not adopted as the filter form (see Per-subject stimulus-space filter), the R+C parameter fit at hV4 under L_{fit} is retained as a per-subject etiological diagnostic. For sub-08, the R+C decomposition recovers $\Delta\lambda = 2.5$ nm (near-zero retinal shift) with $g = -2.25$ (large cortical overshoot), consistent with a cortical-dominant CVD. For sub-09, the R+C decomposition recovers $\Delta\lambda = 19.5$ nm (within the published protan cone-shift range) with $g = -1.10$ (near-physiological cortical compensation, no overshoot), consistent with a retinal-dominant CVD. This per-subject cortical-vs-retinal asymmetry is reported as a mechanistic interpretation of the 2-component fit, not as an independent test.

A.2 Inverse-mapping (pre-image) behavior

The angular form of the 2-component model is bijective by construction: a stimulus-space pre-image $\theta + \delta\theta$ exists for any target hue. The 1-DOF Machado fit for sub-09, by contrast, collapses three hues (green 135° , cyan 180° , blue 225°) onto a single pre-image angle ($\sim 127^\circ$), which mathematically prevents exact inversion at those positions. This arc-compression failure is a structural property of the 1-DOF retinal-shift model and is independent of the fitting criterion; it disqualifies the Machado baseline for the stimulus-space filter task of Section .

A.3 R+C model

The R+C model is retained here as a 2-DOF baseline. Because the cortical gain term g acts as a uniform amplitude scaling without generating the angular distortion captured by the 2-component model's $\beta_c \cos(h - \theta_{\text{conf}})$ term, R+C does not match the observed hV4 LOCO vulnerability pattern beyond what the retinal $\Delta\lambda$ term alone provides; this conclusion is consistent across loss criteria.

Supplementary Methods

S1. Confound Regression and Temporal Filtering

Within-run head motion was estimated via rigid-body realignment. No temporal filtering or confound regression was applied; slow drift was modeled via linear per-run regressors in the general linear model.

S2. Quality Control

Registration quality was evaluated by computing the intersection between the Wang Atlas ROIs (V1, V2, V3, hV4) and the BOLD brain mask in MNI space. Across 10 subjects, mean ROI coverage was 84.3% (SD = 21.7%), and GLM valid ratio (percentage of ROI voxels with reliable stimulus-evoked responses) was 99.6%. We evaluated all subjects' data suitable for downstream analysis, though sub-07 showed reduced coverage (30.8%) due to individual anatomical variability.

S3. Voxel Counts Range

Voxel counts after ROI definition and quality filtering are reported in the main Methods section. Detailed per-subject counts are available in the online repository.

S4. Consistency Test: PCA and Procrustes (TBA)

Additional validation analyzes comparing SRM with PCA-based dimensionality reduction and norm-constrained Procrustes alignment will be reported following completion of the full dataset.

S5. Mean Activation Analysis

We conducted mean voxel response amplitudes analysis to identify whether SRM-based differences rely on the activation. We checked mean amplitudes across all colors and for each color.

S6. Statistical Analysis

All tests used a significance threshold of $\alpha = 0.05$ with False Discovery Rate correction (Benjamini & Hochberg, 1995); $q = 0.05$ applied to RDM color-pair comparisons (28 pairs per participant). Individual CVD participants were compared to the HC distribution using Crawford and Howell (1998) modified t-tests:

$$t^* = \frac{x_{\text{CVD}} - \bar{x}_{\text{HC}}}{\text{SD}_{\text{HC}} \times \sqrt{(n+1)/n}}, \quad df = n - 1 = 6 \quad (4)$$

with one-tailed p-values testing the directional hypothesis that CVD disparity exceeds HC. Effect sizes were quantified via Hedges' g . Group-level permutation tests (10,000 iterations) were used for HC-CVD comparisons by permuting subject labels; color-level permutation tests (1,000 iterations) were used for LOCO/LORO validation by permuting color labels within subjects.

S7. Dimensionality Selection (K-Selection)

The reduced dimension k for SRM was determined via leave-one-subject-out (LOSO) cross-validation based on within-subject representational dissimilarity matrix (RDM) reliability (Cunningham & Yu, 2014). For each candidate k (range: 1–8) and fold, reliability was quantified as the Spearman correlation between RDMs computed from even runs (2,4,6) versus odd runs (1,3,5) of the held-out participant in SRM space. We selected k maximizing mean reliability across seven folds. The selected values and their associated cross-subject RDM Spearman correlations were:

- V1: $k = 4$ (cross-subject RDM correlation = 0.60)
- V2: $k = 4$ (cross-subject RDM correlation = 0.57)
- V3: $k = 3$ (cross-subject RDM correlation = 0.55)
- hV4: $k = 3$ (cross-subject RDM correlation = 0.32)

S8. Generalized Cross-Validation for Ridge Regression

The regularization parameter α for ridge regression was selected via Generalized Cross-Validation (GCV; Golub et al., 1979), implemented as `sklearn.linear_model.RidgeCV` with `gcv_mode='auto'` (Pedregosa et al., 2011). GCV provides a computationally efficient approximation to leave-one-out cross-validation for selecting α . The optimal α minimizes the GCV score:

$$\text{GCV}(\alpha) = \frac{1}{n} \|y - \hat{y}\|_2^2 / \left(1 - \frac{\text{tr}(H)}{n}\right)^2 \quad (5)$$

where X denotes the design matrix, y is the observed response vector, $\hat{y}$ is the ridge-predicted response, $H = X(X^T X + \alpha I)^{-1} X^T$ is the hat matrix, and n is the number of training samples.

S9. Leave-One-Out Consistent Disparity Estimation

To quantify HC-CVD disparity with unbiased references, we used a leave-one-out (LOO) consistent procedure: for each fold i (leaving out HC participant i), a reference pattern R_{-i} was computed as the mean of the remaining six HC participants' aligned data. Both the held-out HC_i and both CVD participants were compared against this same reference R_{-i} , yielding disparity scores $d(\text{HC}_i, R_{-i})$ and $d(\text{CVD}_j, R_{-i})$. This ensures HC and CVD disparities are measured against identical references within each fold. Individual CVD disparity significance was assessed via Crawford and Howell (1998) modified t-tests comparing each CVD participant's mean disparity (averaged across 7 folds) to the HC distribution ($n = 7$, $df = 6$).

S10. Cross-Validation Procedures

Leave-One-Run-Out (LORO)

LORO evaluated whether individual colors could be distinguished from voxel response amplitudes, testing the consistency of neural color encoding across runs. For each of six runs, the corresponding response matrices (B , C) were held out as the test

set, W was trained on the remaining five runs. Both color decoding and voxel response prediction were evaluated on the held-out run.

Leave-One-Color-Out (LOCO)

LOCO tested whether geometric relationships of circular hue space were preserved in neural representation by evaluating interpolation to held-out colors. For each color, all six run response matrices of the target color were held out. W was trained on the remaining seven colors, and the model predicted either the held-out color from voxel responses or voxel responses from the held-out color’s channel outputs.

Leave-One-Subject-Out (LOSO)

LOSO evaluated zero-shot generalization to held-out participants. As LOSO requires cross-subject prediction, SRM-transformed data were used to align voxel spaces across participants. For each HC participant, W was trained on the remaining HC participants’ shared-space data and tested on the held-out participant. This assessed whether the HC group-derived common space supported generalization to novel individuals.

S11. Evaluation Metrics

Color Decoding Metrics

Color decoding was evaluated using (1) mean absolute error (MAE) between predicted and true hue angles (chance = 90° , expected from a uniform random predictor on a circle), and (2) categorization accuracy with $\pm 22.5^\circ$ bins around each of the eight colors (chance = $12.5\% = 1/8$).

Voxel Prediction Metrics

Voxel response prediction was evaluated using Pearson correlation between predicted and observed response vectors, computed across all voxels and colors in the test set.

Statistical Significance

Statistical significance of group comparisons (HC vs CVD) and individual CVD analyzes was assessed using nonparametric permutation tests (10,000 iterations; Nichols & Holmes, 2002).

S12. Shared Response Model Residuals

Each participant’s residuals in the SRM model reflect the extent to which their data are not captured by the common structure, including inherent geometric differences. The orthogonal constraint ($W_i^T W_i = I_k$) helps preserve geometric relationships between colors—distances and angles—in the common space.

Additionally, global between-subject RDM similarity was quantified using Spearman correlation across all 28 color pairs.

S13. Image Orientation Initialization

Image orientation was first initialized according to the scanner-defined obliquity information in the NIfTI header. The mutual-information cost function was then optimized to maximize the statistical dependence between T1w and BOLD intensity distribution within the overlapping region.

S14. Procrustes Alignment Benefits

This preprocessing reduces measurement noise across runs while preserving color representational geometry, thereby improving the quality of subsequent SRM fitting.

S16. Comparison with Retinal-Family Distortion Models

Two retinal-family models were evaluated alongside the 2-component model by the same grid-search and permutation procedure (see Section).

Models. The 1-DOF Machado model Machado et al., 2009 shifts the L/M cone spectral sensitivity by $\Delta\lambda$ nm, yielding a hue-confusion pattern characteristic of the CVD subtype. The 2-DOF retinal+cortical R+C model Tregillus et al., 2021 augments

Machado with a cortical gain g : $r'_g = r_{g,\text{base}} + (1 + g)(r_{g,\text{ret}} - r_{g,\text{base}})$, where $g = 0$ recovers the retinal-only model and $g = -1$ is exact cortical compensation.

Fits (Table 1). Under the historical LOCO- ρ argmax criterion (40,320 label permutations), reported here for reference only, retinal-family fits at hV4 yield Sub-08 R+C ($\Delta\lambda = 2.0$ nm, $g = +2.25$, $\rho = 0.857$) and Sub-09 1-DOF Machado ($\Delta\lambda = 13.5$ nm, $\rho = 0.762$); the latter shift magnitude falls within the established range for moderate–severe protanomaly (Neitz & Neitz, 2011). The large positive cortical gain for Sub-08 ($g = +2.25$, indicating >100% overshoot beyond the exact Tregillus compensation level of $g = -1$; Tregillus et al., 2021) is not physiologically interpretable as a gain parameter and is treated as an empirical characterization of residual hV4 structure. The cross-subject model-class divergence (R+C for Sub-08; 1-DOF Machado for Sub-09) prevents a common-axis comparison of the two cases’ distortion mechanisms.

Correction-vector divergence. Despite comparable LOCO fit quality ($\rho = 0.857$ for R+C vs. 0.881 for 2-component), the R+C and 2-component correction vectors for Sub-08 have cosine similarity -0.18 and sign agreement only 3/8, confirming that detection converges while correction directions diverge at the cortical level.

Δ RDM cross-model convergence. Using R+C parameters for Sub-09 ($\Delta\lambda = 19.5$ nm, $g = -1.10$), an independent Δ RDM test reaches $p = 0.026$ (40,320 cosine-similarity permutations on crossnobis distances), providing cross-model corroboration for the Sub-09 distortion signal. Retinal-family optimization directly against Δ RDM fails for both CVD subjects; LOCO is the more discriminating phenotype and Δ RDM serves as a convergence cross-check only.

Arc-compression and LOCO vulnerability. At Sub-09’s severity, Machado arc-compression ($\Delta\lambda = 13.5$ nm) collapses three hues (c4 green, c5 cyan, c6 blue) to a single perceived angle in opponent hue space. These three collapsed hues fall within Sub-09’s LOCO vulnerability cluster (c5/c6 cyan/blue), confirming that arc-compression eliminates correction capacity precisely where Sub-09 requires it most.

S17. HC leave-one-out magnitude-anchor distribution

Each of $n = 6$ effective HC participants was refit under the canonical V4 LOCO loss L_{fit} (Equation 3) with the remaining HC pool as reference, yielding an HC leave-one-out distribution of fitted-parameter magnitudes $\|\hat{\beta}\|$ ranging 26.3° to 49.2° (mean 40.1°). Sub-07 was excluded because its hV4 ROI contained only 16 voxels (versus 67–70 for the remaining HC), which precluded a stable forward-encoder LOCO target vector at this ROI; the exclusion arises in filter-fit space and is unrelated to the SRM dimensionality choice. This range is reported as a magnitude anchor for the CVD estimates (sub-08 $\|\hat{\beta}\| = 40.5^\circ$, falling near the HC mean; sub-09 $\|\hat{\beta}\| = 22.8^\circ$, below the HC minimum), not as a hypothesis test of CVD-versus-HC specificity.

S18. Effect Sizes for Single-Case Comparisons

Crawford–Howell single-case comparisons used the modified t Crawford & Howell, 1998; the corresponding case-control effect size is $d_{cc} = (x_{\text{case}} - \bar{x}_{\text{HC}})/s_{\text{HC}} = t \cdot \sqrt{(n+1)/n}$ where n is the HC sample size used in the test. Mann–Whitney U tests are reported with rank-biserial correlation r_{rb} as effect size. Full per-test statistics (t , U , p , d_{cc} , r_{rb}) for every Crawford–Howell and Mann–Whitney comparison reported in the main text are tabulated in the online repository (`statistics/effect_sizes.csv`).

Table 1

All distortion model fits at hV4 LOCO for both CVD participants. Retinal-family fits are reported under the historical LOCO- ρ argmax criterion (Section A.1 of Appendix A); ρ is Spearman correlation between simulated and observed per-color vulnerability.

Subject	Model	DOF	Param. 1	Param. 2	ρ
Sub-08 (deutan)	2-component	2	$\beta_s = 38^\circ$	$\beta_c = -14^\circ$	0.881
Sub-09 (protan)	2-component	2	$\beta_s = 6^\circ$	$\beta_c = -22^\circ$	0.690
Sub-08 (deutan)	R+C	2	$\Delta\lambda = 2.0$ nm	$g = +2.25$	0.857
Sub-09 (protan)	Machado	1	$\Delta\lambda = 13.5$ nm	—	0.762

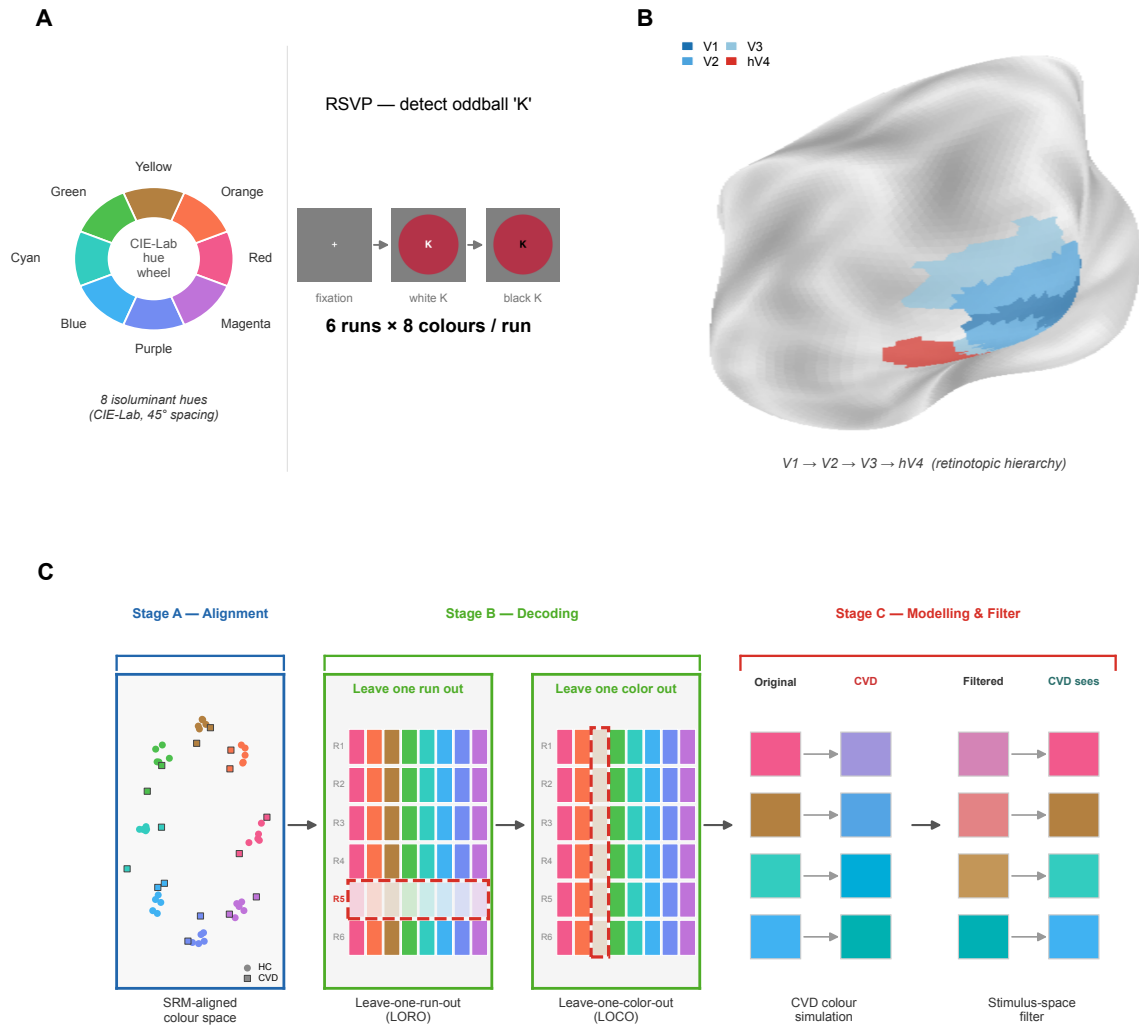


Figure 1. Experimental paradigm and analysis pipeline. (A) Stimulus set: 8 isoluminant CIE $L^*a^*b^*$ hues at 45° angular spacing ($L^* = 75$, chroma = 40; red–magenta). *(B)* Retinotopic ROIs: V1, V2, V3, hV4 (standard retinotopic mapping). *(C)* Analysis pipeline. Stage A: two-stage GLM single-trial β amplitudes, Procrustes-aligned across runs, projected into an SRM space trained on HC participants only. Decoding: LORO (discrimination), LOCO (interpolation). CVD characterization: 2-component model with retinal (β_s) and cortical (β_c) components. Pre-image yields per-subject stimulus-space filter. HC: $n = 7$ (sub-01–07); deutan: sub-08; protan: sub-09.

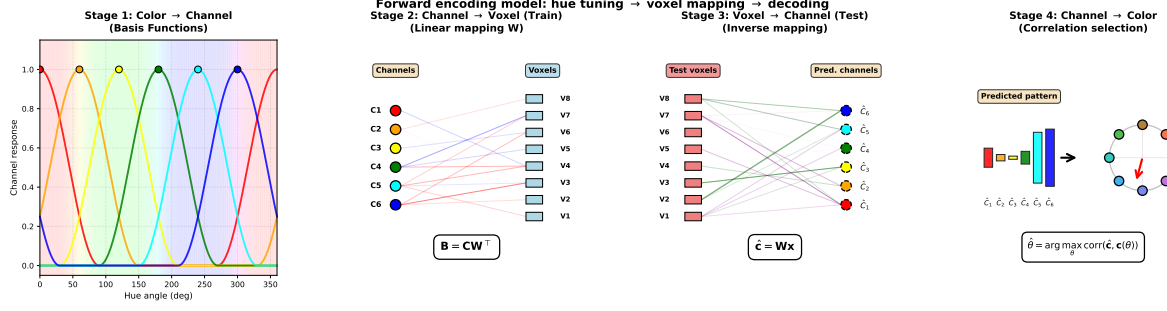


Figure 2. Forward encoding model schematic. Stage 1: each displayed hue is encoded as channel activations by six half-wave rectified squared-cosine basis functions, with preferred hues spaced at 60 degrees (Brouwer & Heeger 2009). Stage 2 (training): a per-voxel weight matrix W is fit by ridge regression with GCV-selected regularization, mapping channels to voxel responses. Stage 3 (testing): for a held-out voxel pattern, channel activations are reconstructed as $\hat{c} = Wx$. Stage 4: the decoded hue is recovered as the angle whose basis vector has the highest Pearson correlation with the reconstructed channel pattern. The same encoder is used for cross-run color-discrimination decoding (Section) and cross-color interpolation decoding (Section); the test set differs but the basis, weights, and correlation-based selection are identical.

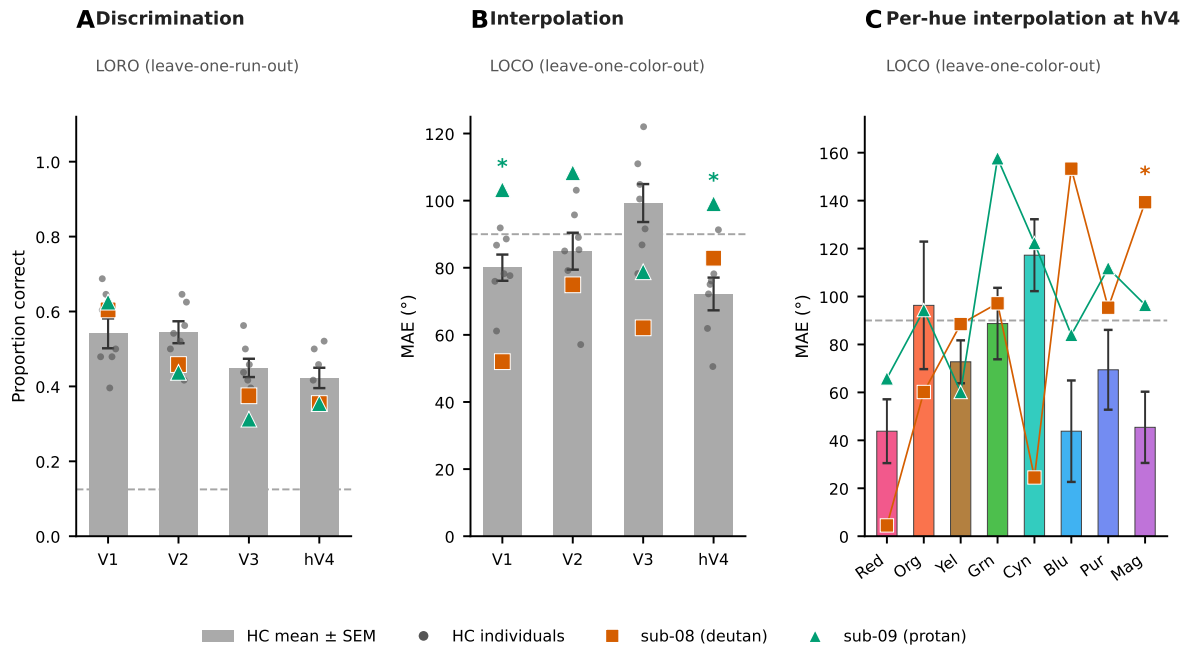


Figure 3. Color discrimination is preserved but interpolation is selectively impaired at hV4 in CVD. (A) LORO discrimination accuracy (forward encoding model, 8-class, SRM-aligned), V1–hV4. HC mean $\pm$ SEM ($n = 7$, gray); sub-08 (orange square), sub-09 (teal triangle). Chance = $1/8$. *(B)* LOCO adjacent accuracy (± 1 hue step; chance = $3/8$). *: sub-09 below HC at hV4 (Crawford & Howell $t = -2.48$, $p = 0.024$; exploratory single-case finding); sub-08 $t = -1.58$, $p = 0.082$, n.s. *(C)* Per-hue adjacent accuracy at hV4, HC bars $\pm$ SEM. Per-hue Crawford & Howell tests (one-tailed): blue — sub-08 $d = 2.13$, $p = 0.047$; sub-09 $d = 2.15$, $p = 0.046$; purple — sub-08 $d = 2.40$, $p = 0.033$. Joint pattern in (A) and (B–C) defines the filter precondition and target, respectively.

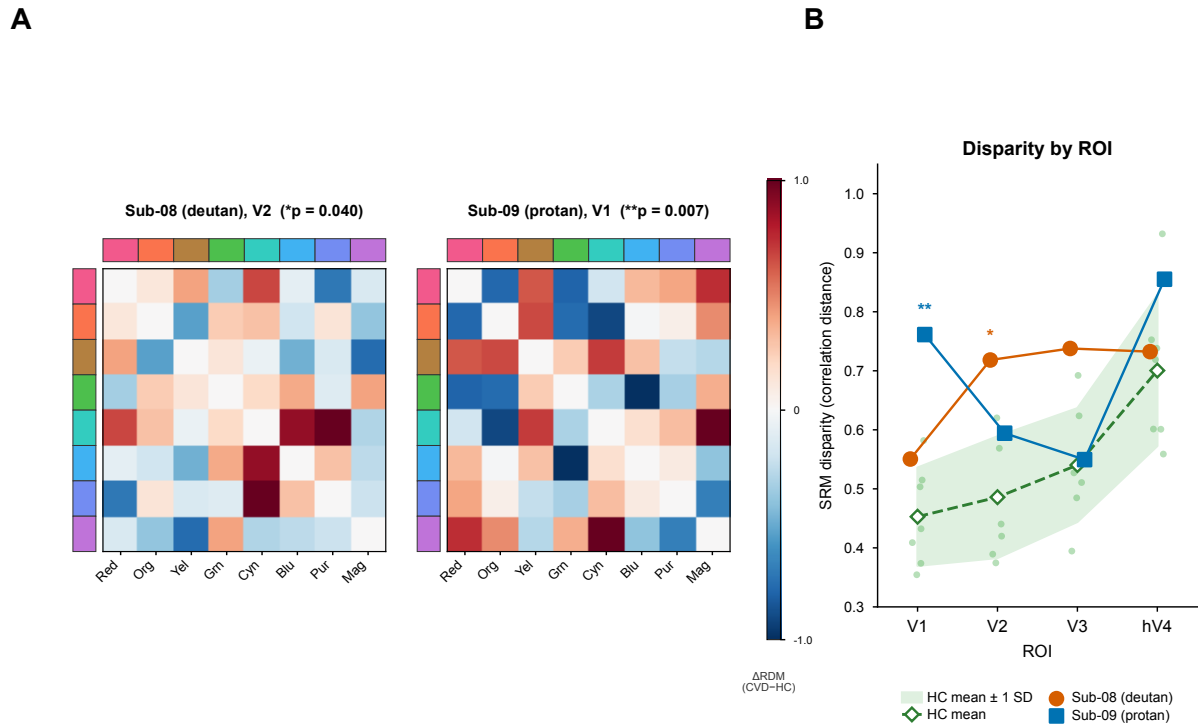


Figure 4. Each CVD subject shows significantly elevated color representation disparity at a distinct ROI. (A) $\Delta RDM = RDM_{CVD} - \overline{RDM}_{HC}$ in SRM-aligned space at each subject's primary ROI (sub-08 V2, sub-09 V1). Warm: CVD distances larger than HC mean; cool: smaller. **(B)** Mean pairwise correlation distance (disparity) per subject per ROI. HC band: group mean $\pm$ 1 SD ($n = 7$); individual HC LOO as dots. Crawford & Howell tests (sub-08 V2: $p = 0.040^*$; sub-09 V1: $p = 0.007^{**}$, exploratory single-case). (A) shows the directional pattern of pairwise deviations. (B) tests the absolute amplitude of pairwise distances against the HC LOO distribution. Idiosyncratic ROI specificity (V2 deutan; V1 protan) is inconsistent with a shared group-level gain mechanism.

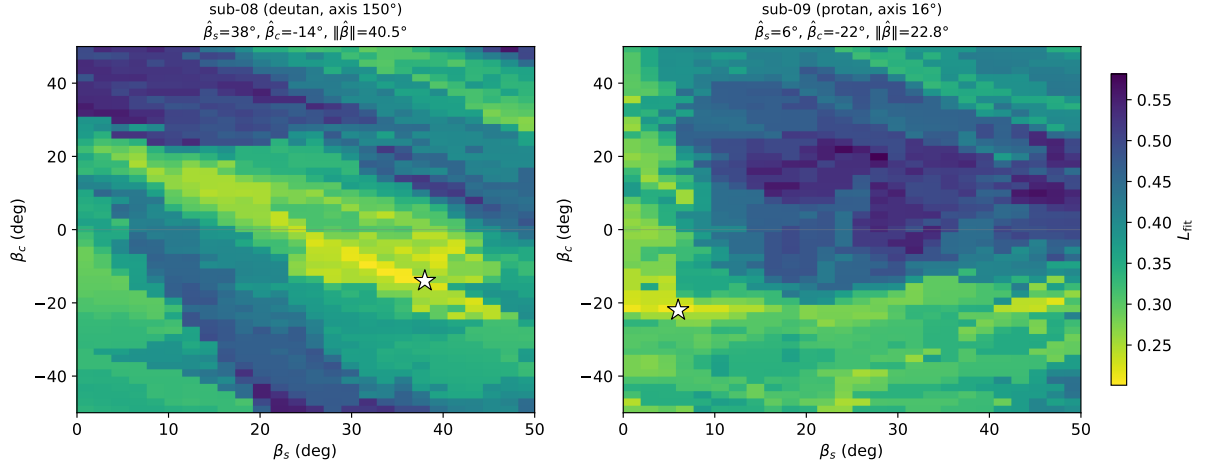


Figure 5. Per-subject 2-component loss landscape at hV4. Canonical V4 LOCO

loss $L_{\text{fit}}(\beta_s, \beta_c) = 1.0 L_{\text{vuln}} + 0.5 L_{\text{rank}} + 0.2 L_{\text{rdm}} + 0.1 L_{\text{smooth}}$ (Equation 3; Methods) evaluated on the $26 \times 51 = 1,326$ -cell grid ($\beta_s \in [0^\circ, 50^\circ]$, $\beta_c \in [-50^\circ, +50^\circ]$, step 2°).

Left: sub-08 (deutan, Stockman S-cone confusion axis 150°). Right: sub-09 (protan, Stockman axis 16°). White star: per-subject argmin — sub-08 ($\hat{\beta}_s, \hat{\beta}_c$) = $(38^\circ, -14^\circ)$, $\|\hat{\beta}\| = 40.5^\circ$; sub-09 ($6^\circ, -22^\circ$), $\|\hat{\beta}\| = 22.8^\circ$. Both $\hat{\beta}_c$ values are negative. The inter-subject asymmetry resides in the S-cone-axis magnitude ($\hat{\beta}_s = 38^\circ$ vs. 6°). See Appendix for the historical LOCO- ρ comparison and Machado/R+C cone-shift baselines.

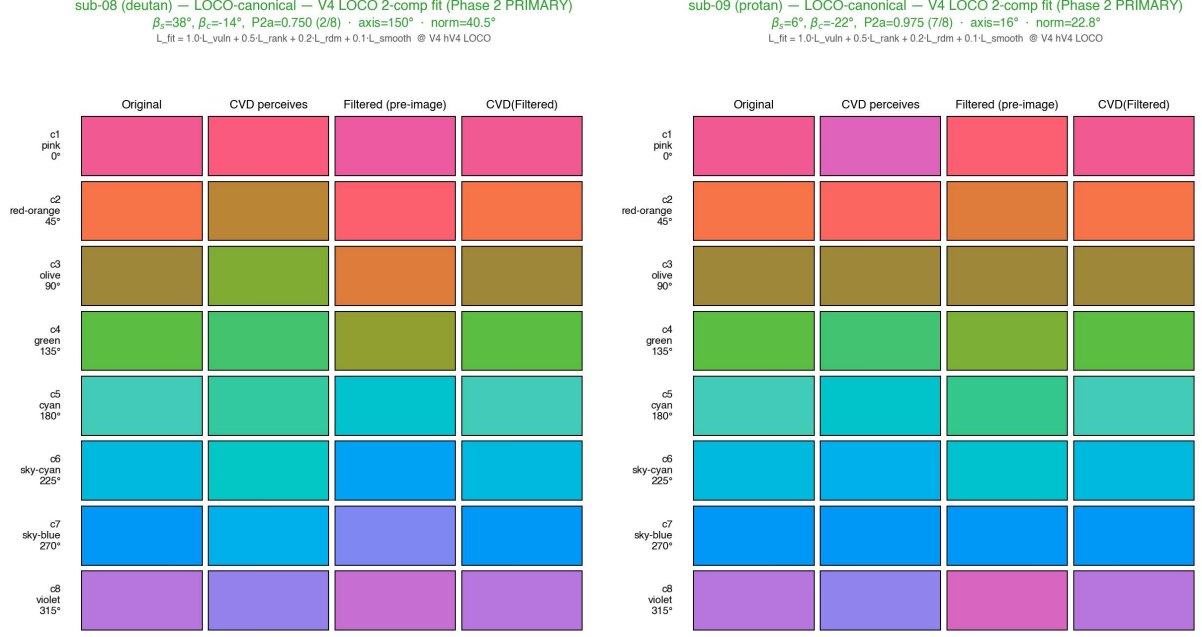


Figure 6. Per-subject stimulus-space filter: 4-column rendering, side-by-side.

Left block: sub-08 (deutan), filter at $(\hat{\beta}_s, \hat{\beta}_c) = (38^\circ, -14^\circ)$, $\|\hat{\beta}\| = 40.5^\circ$. Right block: sub-09 (protan), filter at $(\hat{\beta}_s, \hat{\beta}_c) = (6^\circ, -22^\circ)$, $\|\hat{\beta}\| = 22.8^\circ$. Eight rows correspond to the eight displayed DKL hues. Columns (left to right) within each block: *Original* (HC percept of the displayed stimulus); *CVD percept* (simulated CVD percept of the original under the per-subject 2-component model); *Filtered* (stimulus after applying $\delta\theta^{\text{filt}}$, the exact pre-image of $-\delta\theta(\hat{\beta}_s, \hat{\beta}_c)$ computed by Brent's-method refinement of the forward map; 8/8 exact, residual $< 0.001^\circ$ for both subjects); *CVD(Filt.)* (simulated CVD percept of the filtered stimulus). Rendering coordinate space: STIM_LAB CIELab (project convention). Quantitative behavioral validation of filter efficacy is deferred to the Phase-3 2AFC arm and will be added once data are collected.